66degrees

Overview

In preparation for your next interview, we would like you to complete the following prompt and come prepared to discuss and talk through your approach in a role-play presentation based on the below tasks and deliverables.

The time slot is 45 minutes and you will be meeting with your project team and manager. You may choose any way that you want to communicate your response to this prompt; feel free to use a presentation or diagramming tool of your choice or the provided presentation template. You will present via Google Meet.

The Scenario

The objective of this project is to design and implement a data pipeline that extracts data from Kaggle, transforms it into dimension and fact tables, and loads it into a SQLite database. Additionally, a report will need to be generated based on the dimension and fact tables created. Lastly, draft a solution for deploying this data pipeline and report generation process to the cloud.

Tasks

1. Data Extraction:
   - Use Python for data processing.
     1. Pick any framework.
   - Extract data from:
     1. https://www.kaggle.com/datasets/lovishbansal123/sales-of-a-supermarket
     2. Use the Kaggle API Python package to pull the data:
        1. https://github.com/Kaggle/kaggle-api
2. Schema Design:
   - Identify two dimensions from the dataset.
   - Define the schema for the two dimension tables and one fact table.
3. Transform and Load Data:
   - Transform the extracted data into two dimension tables and one fact table.
   - Load the transformed data into a SQLite database.
4. Reporting:
   - Generate an analytical report by aggregating the data from the dimension and fact tables you created. Solution must include:
     1. Expected output is a SQL Query and the resulting dataset

ii. Should include joins to your dimensions as well as windowing functions

5. Draft a Solution for Cloud Deployment:
    - Design a cloud solution to fully automate the data pipeline and report generation process.
    - Identify the necessary cloud components.
        i. Assume you will not be using SQLite in the cloud.
    - Create an architecture diagram to represent the entire workflow and be prepared to discuss.

Deliverables

- Jupyter notebook or python scripts that encapsulate the tasks above
- Code should be loaded into a git repository and shared with provided users
- Python code for data extraction, transformation, and loading.
- SQL script(s) for creating the tables.
- SQL script for report generation.
- Cloud architecture diagram representing a solution for deploying the data pipeline and report generation process to the cloud.

Can you come to the meeting with diagrams, an approach, and anything else you can think of necessary to share with the team?